(U) Interface Control Document

(U) T-150 UAS AFSIM Plugin Suite



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1 (U) Introduction

1.1 (U) Purpose

(U) This Interface Control Document (ICD) defines the software interfaces for the T-150 UAS AFSIM Plugin Suite. It specifies how each plugin communicates with the AFSIM simulation framework and with other plugins in the suite. The document serves as the authoritative reference for interface agreements between plugin components.

(U) This document is adapted from the NGIC ICD template (originally designed for ODESSA DLL interfaces) to the AFSIM plugin architecture. The following mapping conventions apply:

- (U) "DLL" maps to "Plugin (.dll/.so)"
- (U) "Structs" maps to "AuxData Key-Value Interface"
- (U) "DLL Functions" maps to "Plugin API Entry Points"
- (U) Sections specific to RF, antenna patterns, or frequency interfaces are marked "Not Applicable"

1.2 (U) Scope

(U) This ICD covers the five plugins comprising the T-150 UAS simulation capability:

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(U) **UNCLASSIFIED**

The document covers plugin loading, API entry points, inter-plugin data exchange via the AuxData key-value store, scenario configuration interfaces, and external output interfaces.

1.3 (U) Document Organization

- (U) Section 2 describes how AFSIM discovers, loads, and initializes plugins.
- (U) Section 3 specifies the required C-linkage entry points and registration classes.
- (U) Section 4 defines the complete AuxData key-value interface organized by producer.
- (U) Section 5 describes inter-plugin data flow and producer-consumer relationships.
- (U) Section 6 documents the scenario script configuration interface.
- (U) Section 7 covers external interfaces including file output and visualization.
- (U) Sections 8 and 9 provide references and acronyms.

1.4 (U) Applicable Standards

- (U) AFSIM Plugin API (version determined at compile time via WSF_PLUGIN_API_MAJOR_VERSION / WSF_PLUGIN_API_MINOR_VERSION)
- (U) C++17 language standard
- (U) CMake 3.16+ build system

2 (U) Plugin Interface Overview

2.1 (U) Plugin Discovery and Loading

(U) AFSIM discovers plugins by scanning shared library files in its plugin directories. On Linux, plugins are .so files; on Windows, .dll files. AFSIM loads each library and invokes standardized C-linkage entry points to negotiate version compatibility and register extensions.

2.2 (U) Plugin Directories

(U) Plugins must be installed to both AFSIM system directories to be discovered:

Table 1: (U) Plugin Directories

UNCLASSIFIED	
Directory	Purpose
<AFSIM_ROOT>/lib/wsf_plugins/	Primary plugin search path
<AFSIM_ROOT>/bin/mission_plugins/	Secondary plugin search path
	UNCLASSIFIED

(U) On Linux, the typical AFSIM root is /opt/afsim*/. On Windows, it is the AFSIM installation directory. Plugins placed in project-local directories (e.g., ./lib/) are **not** loaded by the AFSIM runtime.

2.3 (U) Plugin Lifecycle

(U) Each plugin follows a five-stage lifecycle:

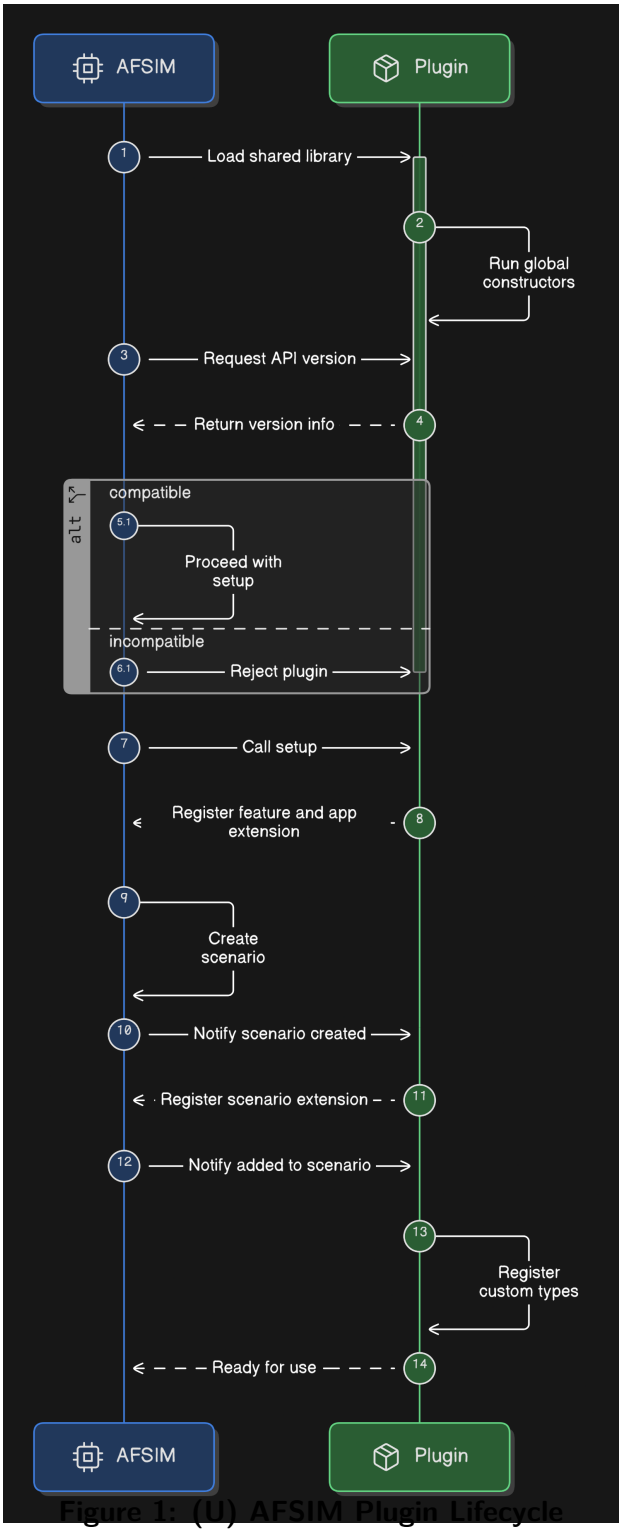


Figure 1: (U) AFSIM Plugin Lifecycle

2.4 (U) Feature Registration

(U) Each plugin registers a feature name that scenario files can query to verify plugin availability:

Table 2: (U) Feature Registration

UNCLASSIFIED		
Plugin	Feature Name	Extension Name
t150_battery	t150_battery	t150_battery_registration
t150_propulsion	t150_propulsion	t150_propulsion_registration
t150_vrs	t150_vrs	t150_vrs_registration
t150_slung_load	t150_slung_load	t150_slung_load_registration
rotorcraft_6dof	rotorcraft_6dof	rotorcraft_6dof_registration
		UNCLASSIFIED

3 (U) Plugin API Entry Points

3.1 (U) Common Entry Points

(U) All five plugins export the same two C-linkage functions. The function signatures use AFSIM-provided types.

3.1.1 (U) WsfPluginVersion

```
1 extern "C" EXPORT_MACRO void WsfPluginVersion(UtPluginVersion& aVersion)
```

(U) **Purpose:** Returns the AFSIM API version the plugin was compiled against.

(U) **Behavior:** Constructs a UtPluginVersion from compile-time constants:

Table 3: (U) WsfPluginVersion Constants

UNCLASSIFIED	
Constant	Description
WSF_PLUGIN_API_MAJOR_VERSION	AFSIM major API version
WSF_PLUGIN_API_MINOR_VERSION	AFSIM minor API version
WSF_PLUGIN_API_SYSTEMABI_STRING	Platform ABI identifier
	UNCLASSIFIED

(U) AFSIM compares these values against its own version. If the major version differs, the plugin is rejected.

3.1.2 (U) WsfPluginSetup

```
1 extern "C" EXPORT_MACRO void WsfPluginSetup(WsfApplication& aApplication)
```

(U) **Purpose:** Registers the plugin's feature name and application extension.

(U) **Behavior:**

1. (U) Checks aApplication.ExtensionIsRegistered() to prevent duplicate registration.
2. (U) Calls aApplication.RegisterFeature() with the plugin's feature name.
3. (U) Calls aApplication.RegisterExtension() with a new ApplicationExtension instance.

(U) The battery plugin additionally prints a CUI banner on first plugin load (coordinated via the NGIC_BANNER_PRINTED environment variable to print only once across all NGIC plugins).

3.2 (U) ApplicationExtension::ScenarioCreated

```
void ScenarioCreated(WsfScenario& aScenario) override
```

(U) **Purpose:** Registers a ScenarioExtension with the newly created scenario.

(U) **Behavior:** Calls aScenario.RegisterExtension() with a scenario-specific registration object. Most plugins use WsfDefaultApplicationExtension<RegistrationType> as a convenience wrapper. The battery plugin uses a custom ApplicationExtension class.

3.3 (U) ScenarioExtension::AddedToScenario

```
void AddedToScenario() override
```

(U) **Purpose:** Registers the plugin's processor or mover type so that scenario files can instantiate it.

(U) **Behavior by plugin:**

Table 4: (U) Type Registration

UNCLASSIFIED		
Plugin	Registration Call	Type Name
t150_battery	WsfProcessorTypes::Get(scen).AddCoreType(...)	T150_BATTERY_PROCESSOR
t150_propulsion	WsfProcessorTypes::Get(scen).AddCoreType(...)	T150_PROPULSION_PROCESSOR
t150_vrs	WsfProcessorTypes::Get(scen).AddCoreType(...)	T150_VRS_PROCESSOR
t150_slung_load	WsfProcessorTypes::Get(scen).AddCoreType(...)	T150_SLUNG_LOAD_PROCESSOR
rotorcraft_6dof	WsfMoverTypes::Get(scen).AddCoreType(...)	WSF_ROTORCRAFT_6DOF_MOVER
		UNCLASSIFIED

(U) Each AddCoreType call provides a prototype instance that AFSIM clones (via the Clone() method) for each platform that uses that type.

4 (U) AuxData Interface

(U) The AuxData interface is the primary mechanism for inter-plugin communication. Each plugin writes key-value pairs to the platform's WsfAttributeContainer (accessed via GetPlatform()->GetAuxData()). Other plugins and scenario scripts can read these values. All keys are string identifiers; values are typed as double, int, bool, or string.

4.1 (U) Battery Processor Outputs (9 Keys)

(U) Producer: T150_BATTERY_PROCESSOR

Table 5: (U) Battery Processor AuxData Keys

UNCLASSIFIED				
AuxData Key	Type	Units	Range	Description
BATTERY_SOC	double	–	0.0–1.0	State of charge (1.0 = fully charged)
BATTERY_POWER_W	double	W	≥ 0	Current total electrical power draw
BATTERY_ENERGY_WH	double	Wh	≥ 0	Remaining energy in battery
BATTERY_VOLTAGE	double	V	39.0–54.6	Terminal voltage under load
BATTERY_OCV	double	V	39.0–54.6	Open-circuit voltage (no-load)
BATTERY_VOLTAGE_SAG	double	V	≥ 0	Voltage drop due to internal resistance
BATTERY_CURRENT_A	double	A	≥ 0	Current draw from battery
BATTERY_LOW	bool	–	true/false	SOC below low-battery threshold (default 20%)
BATTERY_VOLTAGE_LOW	bool	–	true/false	Terminal voltage below sag warning level
				UNCLASSIFIED

(U) **Update rate:** 10 Hz (100 ms update interval)

(U) **Notes:** BATTERY_LOW is set once when SOC drops below threshold and does not reset. BATTERY_VOLTAGE_LOW resets when voltage recovers above the warning level plus 1.0 V hysteresis.

4.2 (U) Propulsion Processor Outputs (19 Keys)

(U) Producer: T150_PROPULSION_PROCESSOR

Table 6: (U) Propulsion Processor AuxData Keys

UNCLASSIFIED				
AuxData Key	Type	Units	Range	Description
PROP_TOTAL_THRUST	double	N	≥ 0	Total thrust from all rotors (with ground effect)
PROP_TOTAL_POWER	double	W	≥ 0	Total electrical power consumption
PROP_SYSTEM_EFF	double	–	0.0–1.0	Overall system efficiency (motor $\times$ coaxial)
PROP_MOTOR_COUNT	int	count	0–8	Number of currently operational motors
PROP_MAX_TEMP	double	°C	-40–200	Hottest motor temperature
PROP_AVG_TEMP	double	°C	-40–200	Average motor temperature
PROP_GROUND_EFFECT	double	–	1.0–1.3	Ground effect thrust multiplier
PROP_ALTITUDE_AGL	double	m	≥ 0	Current altitude above ground level
PROP_TOTAL_MASS	double	kg	> 0	Total mass (empty + payload)
PROP_PAYLOAD_MASS	double	kg	≥ 0	Payload mass
PROP_THRUST_MARGIN	double	–	0.0–1.0	Available thrust margin above hover
PROP_CG_OFFSET_X	double	m	any	Combined CG offset, forward axis
PROP_CG_OFFSET_Y	double	m	any	Combined CG offset, lateral axis
PROP_CG_OFFSET_Z	double	m	any	Combined CG offset, vertical axis
PROP_OVERLOADED	bool	–	true/false	Payload exceeds max capacity
				UNCLASSIFIED

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Table 6: (U) Propulsion Processor AuxData Keys (Continued)

UNCLASSIFIED				
AuxData Key	Type	Units	Range	Description
PROP_ FORWARD_ SPEED	double	m/s	≥ 0	Horizontal ground speed
PROP_CLIMB_ RATE	double	m/s	any	Vertical speed (positive = climbing)
PROP_BANK_ ANGLE	double	deg	0–60	Estimated bank angle
PROP_ACCEL_ MAG	double	m/s^2	≥ 0	Horizontal acceleration magnitude
				UNCLASSIFIED

(U) **Update rate:** 10 Hz (100 ms update interval)

4.3 (U) VRS Processor Outputs (10 Keys)

(U) Producer: T150_VRS_PROCESSOR

Table 7: (U) VRS Processor AuxData Keys

UNCLASSIFIED				
AuxData Key	Type	Units	Range	Description
VRS_MARGIN	double	–	0.0–1.0	Distance from VRS boundary (0 = in VRS, 1 = clear)
VRS_STATE	int	enum	0–4	VRS state machine value (see below)
VRS_LIFT_ FACTOR	double	–	0.4–1.0	Lift degradation factor (1.0 = no degradation)
VRS_CONTROL_ FACTOR	double	–	0.5–1.0	Control authority factor (1.0 = full authority)
VRS_ DEVELOPMENT	double	–	0.0–1.0	VRS development progression
VRS_DESCENT_ RATE	double	m/s	≥ 0	Current descent rate (positive = descending)
VRS_FORWARD_ SPEED	double	m/s	≥ 0	Current horizontal speed
				UNCLASSIFIED

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Table 7: (U) VRS Processor AuxData Keys (Continued)

UNCLASSIFIED				
AuxData Key	Type	Units	Range	Description
VRS_CAUTION	bool	–	true/false	Approaching VRS envelope
VRS_ONSET	bool	–	true/false	VRS onset detected
VRS_CRITICAL	bool	–	true/false	VRS fully developed, recovery action required
				UNCLASSIFIED

(U) **VRS_STATE enum values:**

Table 8: (U) VRS State Enumeration

UNCLASSIFIED	
Value	Description
0	CLEAR – No VRS risk
1	CAUTION – Approaching VRS boundary
2	ONSET – VRS conditions beginning
3	DEVELOPED – Full VRS, severe lift/control degradation
4	RECOVERY – Transitioning out of VRS
	UNCLASSIFIED

(U) **Update rate:** 10 Hz (100 ms update interval)

4.4 (U) Slung Load Processor Outputs (14 Keys)

(U) Producer: T150_SLUNG_LOAD_PROCESSOR

Table 9: (U) Slung Load Processor AuxData Keys

UNCLASSIFIED				
AuxData Key	Type	Units	Range	Description
SLUNG_LOAD_STATE	int	enum	0–4	Load state (see below)
SLUNG_LOAD_MASS	double	kg	≥ 0	Attached load mass
				UNCLASSIFIED

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Table 9: (U) Slung Load Processor AuxData Keys (Continued)

UNCLASSIFIED				
AuxData Key	Type	Units	Range	Description
SLUNG_CABLE_LENGTH	double	m	≥ 0.5	Cable length
SLUNG_CABLE_TENSION	double	N	≥ 0	Current cable tension
SLUNG_SWING_X	double	deg	-70–70	Pendulum angle in pitch (X-Z plane)
SLUNG_SWING_Y	double	deg	-70–70	Pendulum angle in roll (Y-Z plane)
SLUNG_SWING_MAG	double	deg	0–70	Combined swing angle magnitude
SLUNG_FORCE_X	double	N	any	Cable reaction force on aircraft, forward axis
SLUNG_FORCE_Y	double	N	any	Cable reaction force on aircraft, lateral axis
SLUNG_FORCE_Z	double	N	≥ 0	Cable reaction force on aircraft, vertical axis
SLUNG_POS_X	double	m	any	Load position relative to aircraft CG, forward
SLUNG_POS_Y	double	m	any	Load position relative to aircraft CG, lateral
SLUNG_POS_Z	double	m	any	Load position relative to aircraft CG, vertical
SLUNG_NATURAL_FREQ	double	Hz	> 0	Pendulum natural frequency
SLUNG_RESONANCE	bool	–	true/false	Resonance condition detected
				UNCLASSIFIED

(U) **SLUNG_LOAD_STATE enum values:**

Table 10: (U) Slung Load State Enumeration

UNCLASSIFIED	
Value	Description
0	DETACHED – No load attached
1	ATTACHED – Load attached, stable
	UNCLASSIFIED

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Table 10: (U) Slung Load State Enumeration (Continued)

UNCLASSIFIED	
Value	Description
2	SWINGING – Swing exceeds resonance threshold
3	RESONANCE – Swing frequency matches natural frequency at large amplitude
4	RELEASED – Load released (cable overload or commanded)
	UNCLASSIFIED

(U) **Update rate:** 50 Hz (20 ms update interval, configurable)

4.5 (U) 6DOF Mover Outputs (16 Keys)

(U) Producer: WSF_ROTORCRAFT_6DOF_MOVER

Table 11: (U) 6DOF Mover AuxData Keys

UNCLASSIFIED			
AuxData Key	Type	Units	Description
6dof_pos_n	double	m	Local NED position, North
6dof_pos_e	double	m	Local NED position, East
6dof_pos_d	double	m	Local NED position, Down
6dof_vel_n	double	m/s	NED velocity, North
6dof_vel_e	double	m/s	NED velocity, East
6dof_vel_d	double	m/s	NED velocity, Down (positive = descending)
6dof_roll_deg	double	deg	Roll angle
6dof_pitch_deg	double	deg	Pitch angle
6dof_yaw_deg	double	deg	Yaw angle (heading)
6dof_p_dps	double	deg/s	Roll rate
6dof_q_dps	double	deg/s	Pitch rate
6dof_r_dps	double	deg/s	Yaw rate
6dof_total_thrust	double	N	Total rotor thrust
6dof_total_power	double	W	Total mechanical power (shaft)
6dof_throttle_cmd	double	0–1	Throttle command
6dof_avg_rpm	double	RPM	Average rotor RPM
6dof_mass	double	kg	Current aircraft mass
			UNCLASSIFIED

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Table 11: (U) 6DOF Mover AuxData Keys (Continued)

UNCLASSIFIED			
AuxData Key	Type	Units	Description
6dof_weight	double	N	Current weight force
6dof_target_alt	double	m MSL	Target altitude
			UNCLASSIFIED

(U) **Update rate:** 100 Hz (10 ms update interval)

4.6 (U) 6DOF Mover Consumed Keys

(U) The 6DOF mover reads AuxData keys published by other processors to integrate external forces and moments:

Table 12: (U) 6DOF Mover Consumed AuxData Keys

UNCLASSIFIED		
Consumed Key	Source Plugin	Purpose
SLUNG_FORCE_X/Y/Z	Slung Load Processor	Cable reaction forces on aircraft
WEAPON_FORCE_X/Y/Z	Weapon Processor (legacy)	Weapon recoil forces
WEAPON_<name>_ FORCE_X/Y/Z	Weapon Processor (registry)	Per-weapon recoil forces
WEAPON_MOMENT_X/Y/Z	Weapon Processor (legacy)	Weapon recoil moments
WEAPON_<name>_ MOMENT_X/Y/Z	Weapon Processor (registry)	Per-weapon recoil moments
WEAPON_REGISTRY	Weapon Processor	Comma-separated list of weapon names
WEAPON_<name>_MASS_ DELTA	Weapon Processor	Ammunition mass change per weapon
		UNCLASSIFIED

5 (U) Inter-Plugin Data Flow

5.1 (U) Data Flow Diagram

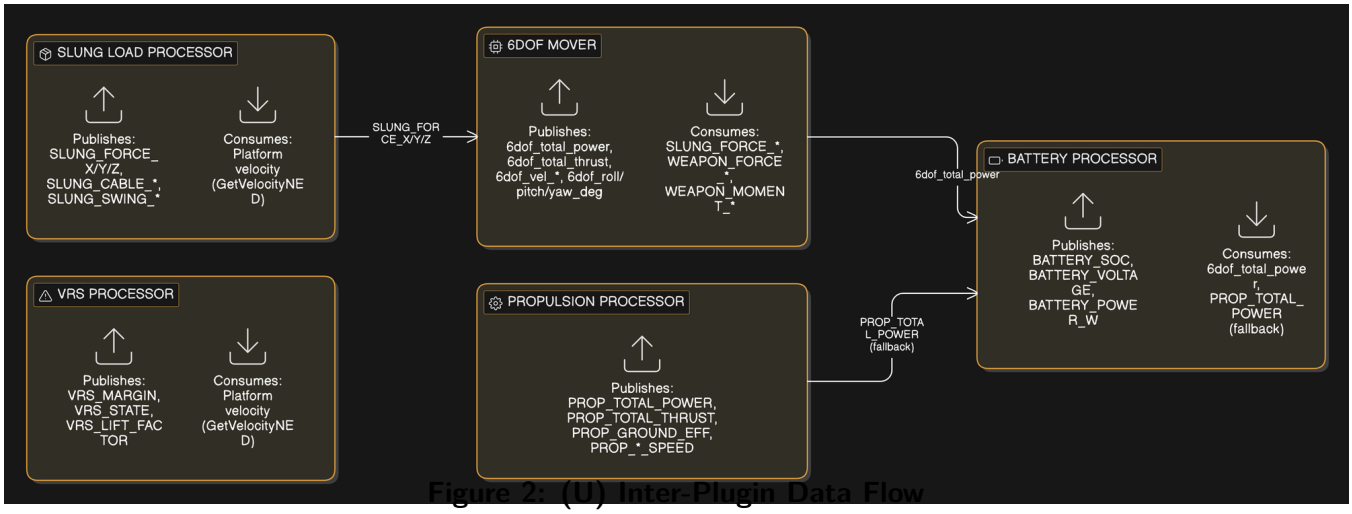


Figure 2: (U) Inter-Plugin Data Flow

5.2 (U) Producer-Consumer Relationships

Table 13: (U) Producer-Consumer Data Paths

UNCLASSIFIED				
Data Path	Producer	Consumer	Key(s)	Direction
Mover power to battery	6DOF Mover	Battery Proc.	6dof_total_power	Mover → Battery
Propulsion power to battery	Propulsion Proc.	Battery Proc.	PROP_TOTAL_POWER	Prop → Battery (fallback)
Slung load forces to mover	Slung Load Proc.	6DOF Mover	SLUNG_FORCE_X/Y/Z	Slung Load → Mover
Platform velocity to VRS	AFSIM Mover	VRS Proc.	GetVelocityNED()	Platform API
Platform velocity to slung load	AFSIM Mover	Slung Load Proc.	GetVelocityNED()	Platform API
				UNCLASSIFIED

5.3 (U) Battery Power Source Fallback Chain

(U) The Battery Processor uses a three-level fallback chain when use_dynamic_power is enabled:

1. (U) **Configured key** – reads from power_source_key (default: PROP_TOTAL_POWER, typically overridden to 6dof_total_power)
2. (U) **6DOF mover power** – reads 6dof_total_power if the configured key returns zero or is absent
3. (U) **Propulsion processor power** – reads PROP_TOTAL_POWER if neither of the above yields a positive value
4. (U) **Fixed fallback** – uses hover_power_w if no dynamic source is available

5.4 (U) Platform API Interfaces

(U) The VRS Processor and Slung Load Processor do not consume AuxData from other plugins. Instead, they read flight state directly from the AFSIM platform API:

- (U) GetPlatform()->GetVelocityNED(velocityNED) – provides NED velocity vector
- (U) GetPlatform()->GetAltitude() – provides altitude MSL

(U) Both processors derive acceleration by numerical differentiation of velocity across timesteps, with acceleration values clamped to filter numerical noise.

6 (U) Scenario Script Interface

6.1 (U) Configuration Mechanism

(U) Each plugin parses its configuration via the ProcessInput(UtInput& aInput) method. AFSIM calls this method for each line within a processor ... end_processor or mover ... end_mover block in the scenario file. The method reads the command keyword via aInput.GetCommand() and the value via aInput.ReadValue() or aInput.ReadBool().

6.2 (U) String Parameter Quote Stripping

(U) AFSIM's ReadValue() for string parameters includes surrounding double quotes in the returned value. Plugins that accept string parameters (e.g., power_source_key, csv_output_file) must strip these quotes. The Battery Processor implements this explicitly:

```
if (mPowerSourceKey.front() == '"' && mPowerSourceKey.back() == '"')
   mPowerSourceKey = mPowerSourceKey.substr(1, mPowerSourceKey.size() - 2);
```

(U) Scenario authors must enclose string values in quotes: power_source_key "6dof_total_power".

6.3 (U) Battery Processor Configuration Parameters

(U) Type name: T150_BATTERY_PROCESSOR

Table 14: (U) Battery Processor Configuration

UNCLASSIFIED				
Parameter	Type	Default	Units	Description
battery_capacity_wh	double	7000.0	Wh	Total battery energy capacity
initial_soc	double	1.0	0–1	Starting state of charge
hover_power_w	double	18000.0	W	Fallback hover power consumption
low_battery_warning	double	0.20	0–1	SOC threshold for low battery warning
nominal_voltage	double	46.8	V	Nominal pack voltage
full_charge_voltage	double	54.6	V	Full charge pack voltage
cutoff_voltage	double	39.0	V	Discharge cutoff voltage
internal_resistance	double	0.025	Ω	Pack internal resistance
voltage_sag_warning	double	42.0	V	Voltage warning threshold
use_dynamic_power	bool	false	–	Enable reading power from AuxData
				UNCLASSIFIED

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Table 14: (U) Battery Processor Configuration (Continued)

UNCLASSIFIED				
Parameter	Type	Default	Units	Description
power_source_key	string	PROP_TOTAL_POWER	–	AuxData key for power input
auxiliary_power_w	double	4000.0	W	Constant auxiliary systems draw
motor_efficiency	double	0.85	0.1–1.0	Mechanical-to-electrical conversion
min_flight_power_factor	double	0.50	0.1–1.0	Min power as fraction of hover
cruise_power_w	double	17000.0	W	Power at cruise speed
cruise_speed_mps	double	30.0	m/s	Cruise speed for velocity model
use_velocity_power_model	bool	true	–	Enable velocity-based power estimation
csv_output_file	string	(empty)	path	CSV telemetry output file path
csv_log_interval	double	1.0	s	Interval between CSV log entries
				UNCLASSIFIED

(U) **Example scenario configuration:**

```
processor battery_monitor T150_BATTERY_PROCESSOR
   battery_capacity_wh 7000
   initial_soc 1.0
   nominal_voltage 46.8
   full_charge_voltage 54.6
   cutoff_voltage 39.0
   internal_resistance 0.025
   use_dynamic_power true
   power_source_key "6dof_total_power"
   hover_power_w 18000
   motor_efficiency 0.37
   auxiliary_power_w 4000
   use_velocity_power_model false
   csv_output_file /path/to/output/battery.csv
   csv_log_interval 1.0
end_processor
```

6.4 (U) Propulsion Processor Configuration Parameters

(U) Type name: T150_PROPULSION_PROCESSOR

Table 15: (U) Propulsion Processor Configuration

UNCLASSIFIED				
Parameter	Type	Default	Units	Description
motor_count	int	8	count	Number of motors (max 8)
kv_rating	double	100.0	RPM/V	Motor Kv rating
max_power_per_motor	double	3500.0	W	Maximum power per motor
propeller_diameter	double	0.76	m	Propeller diameter
coaxial_efficiency	double	0.85	0.5–1.0	Coaxial rotor interference factor
motor_thermal_limit	double	120.0	°C	Motor overtemperature threshold
esc_thermal_limit	double	90.0	°C	ESC overtemperature threshold
nominal_voltage	double	46.8	V	Battery pack nominal voltage
peak_motor_efficiency	double	0.92	0.5–0.98	Peak motor efficiency
ambient_temperature	double	25.0	°C	Ambient temperature
motor_time_constant	double	0.15	s	First-order motor lag time constant
ground_effect_enabled	bool	true	–	Enable ground effect model
rotor_height	double	0.5	m	Rotor height above ground reference
ground_effect_max_boost	double	1.3	1.0–1.5	Max ground effect thrust multiplier
empty_mass	double	68.0	kg	Aircraft empty mass
payload_mass	double	0.0	kg	Payload mass
max_payload_mass	double	55.0	kg	Maximum payload capacity
payload_cg_offset_x/y/z	double	0/0/0.2	m	Payload CG offset from aircraft CG
drag_coefficient	double	1.0	0.1–2.0	Bluff body drag coefficient
frontal_area	double	0.30	m^2	Frontal reference area
induced_power_factor	double	1.15	1.0–1.5	Non-ideal rotor power factor
csv_output_file	string	(empty)	path	CSV telemetry output file path
csv_log_interval	double	1.0	s	Interval between CSV log entries
UNCLASSIFIED				

6.5 (U) VRS Processor Configuration Parameters

(U) Type name: T150_VRS_PROCESSOR

Table 16: (U) VRS Processor Configuration

UNCLASSIFIED				
Parameter	Type	Default	Units	Description
vrs_descent_threshold	double	4.0	m/s	Descent rate for VRS onset
vrs_speed_threshold	double	5.0	m/s	Forward speed that prevents VRS
caution_margin	double	0.8	0.5–0.95	VRS margin level for caution
induced_velocity	double	8.0	m/s	Estimated hover induced velocity
coaxial_factor	double	0.85	0.5–1.0	Coaxial rotor VRS resistance factor
vrs_development_time	double	2.0	s	Time to develop full VRS
vrs_recovery_time	double	3.0	s	Time to recover from VRS
max_lift_degradation	double	0.6	0.0–1.0	Maximum lift loss fraction in VRS
max_control_degradation	double	0.5	0.0–1.0	Maximum control authority loss in VRS
				UNCLASSIFIED

6.6 (U) Slung Load Processor Configuration Parameters

(U) Type name: T150_SLUNG_LOAD_PROCESSOR

Table 17: (U) Slung Load Processor Configuration

UNCLASSIFIED				
Parameter	Type	Default	Units	Description
load_mass	double	0.0	kg	Attached load mass (>0 auto-attaches)
cable_length	double	5.0	m	Cable length (min 0.5 m)
cable_damping	double	0.1	–	Pendulum damping coefficient
attach_point_x/y/z	double	0/0/0.5	m	Cable attach point relative to CG
max_cable_tension	double	5000.0	N	Cable tension limit (triggers release)
resonance_threshold	double	20.0	deg	Swing angle for resonance warning
update_interval	double	0.02	s	Physics update interval (min 1 ms)
				UNCLASSIFIED

6.7 (U) 6DOF Mover Configuration Parameters

(U) Type name: WSF_ROTORCRAFT_6DOF_MOVER

(U) The 6DOF mover inherits route-handling parameters from WsfRouteMover and adds physics configuration:

Table 18: (U) 6DOF Mover Configuration

UNCLASSIFIED				
Parameter	Type	Default	Units	Description
mass	double	–	kg	Total aircraft mass
inertia_xx/yy/zz	double	–	kg·m^2	Moments of inertia
rotor_configuration	string	–	–	COAXIAL_QUAD, COAXIAL_HEX, etc.
rotor_diameter	double	–	m	Rotor/propeller diameter
arm_length	double	–	m	Motor arm length from CG
coaxial_spacing	double	–	m	Vertical spacing between coaxial rotors
thrust_coefficient	double	–	–	Rotor thrust coefficient (Ct)
torque_coefficient	double	–	–	Rotor torque coefficient (Cq)
max_rpm	double	–	RPM	Maximum rotor RPM
motor_time_constant	double	–	s	Motor first-order lag
drag_enabled	bool	–	–	Enable aerodynamic drag model
drag_coefficient_x/y/z	double	–	–	Directional drag coefficients
frontal_area	double	–	m^2	Reference frontal area
ground_effect_enabled	bool	–	–	Enable ground effect model
ground_effect_height_factor	double	–	–	Ground effect height scaling
ground_effect_max_boost	double	–	–	Maximum thrust augmentation
roll/pitch/yaw_mix_gain	double	–	–	Control mixing gains
altitude_hold	bool	–	–	Enable altitude hold controller
altitude_kp/kd	double	–	–	Altitude controller gains
position_hold	bool	–	–	Enable position hold controller
position_kp/kd	double	–	–	Position controller gains
velocity_kp	double	–	–	Velocity controller gain
max_vel_accel	double	–	m/s^2	Maximum velocity acceleration
max_tilt_angle	double	–	deg	Maximum tilt for position control
heading_hold	bool	–	–	Enable heading hold controller
heading_kp	double	–	–	Heading controller gain
waypoint_arrival_radius	double	–	m	Distance to consider waypoint reached
telemetry_enabled	bool	–	–	Enable AuxData telemetry publishing
				UNCLASSIFIED

Continued on next page

Table 18: (U) 6DOF Mover Configuration (Continued)

UNCLASSIFIED				
Parameter	Type	Default	Units	Description
spline_interpolation	bool	false	–	Enable Catmull-Rom spline path
spline_tension	double	–	–	Spline tension parameter
max_deceleration	double	–	m/s^2	Max deceleration for speed profiling
default_waypoint_ behavior	string	FLY_ THROUGH	–	FLY_THROUGH, STOP_AND_HOVER, SLOW_PASS
default_pass_speed	double	–	m/s	Speed for SLOW_PASS behavior
default_hover_ duration	double	–	s	Hover time for STOP_AND_HOVER
				UNCLASSIFIED

7 (U) External Interfaces

7.1 (U) CSV File Output

(U) Both the Battery Processor and Propulsion Processor support CSV telemetry logging when configured with the csv_output_file parameter.

(U) **Battery Processor CSV columns:**

```
Time_s, SOC_pct, Voltage_V, Current_A, Power_W, Energy_Consumed_Wh,
OCV_V, Voltage_Sag_V, Remaining_Energy_Wh
```

(U) **Propulsion Processor CSV columns:**

```
Time_s, Total_Thrust_N, Total_Power_W, System_Efficiency,
Altitude_AGL_m, Ground_Effect_Factor, Total_Mass_kg, Payload_Mass_kg,
Thrust_Margin, Forward_Speed_ms, Climb_Rate_ms, Bank_Angle_deg,
Motor_Avg_RPM, Motor_Avg_Temp_C, Motor_Avg_Current_A
```

(U) The Battery Processor appends a summary block to the CSV at simulation end, including flight duration, initial/final SOC, total energy consumed, and average power.

(U) Log interval is controlled by csv_log_interval (default 1.0 s, minimum 0.1 s).

7.2 (U) AFSIM Event Pipe (AER Output)

(U) All plugins output warnings and informational messages via the AFSIM logging API (ut::log::info(), ut::log::warning()). These messages appear in the AFSIM event log and, when configured, are captured in AER (AFSIM Event Record) output files via the event_pipe scenario directive. AER output captures simulation events including battery warnings, motor overheat events, VRS state transitions, and slung load releases.

7.3 (U) AFSIM Mystic Visualization

(U) All AuxData keys are accessible to the AFSIM Mystic visualization tool for real-time monitoring during simulation. The 6DOF mover additionally synchronizes platform position, velocity, and orientation to the AFSIM platform object via SetLocationLLA(), SetVelocityNED(), and SetOrientationNED(), enabling standard AFSIM 3D visualization.

7.4 (U) Simulation Federation Interfaces

(U) Not Applicable. The T-150 UAS plugin suite does not implement HLA, DIS, or other simulation federation interfaces.

7.5 (U) RF and Antenna Interfaces

(U) Not Applicable. The T-150 UAS plugin suite does not implement RF transmission, antenna pattern, or

frequency management interfaces. Radar detectability is handled through standard AFSIM radar signature definitions in the platform type, not through the plugin interface.

8 (U) References

Table 19: (U) References

UNCLASSIFIED	
Ref	Document
[1]	AFSIM User's Guide, Air Force Research Laboratory
[2]	AFSIM Plugin Developer's Guide
[3]	Malloy Aeronautics T-150 / TRV-150 Technical Specifications
[4]	Johnson, W., *Helicopter Theory*, Dover Publications, 1994
[5]	Leishman, J.G., *Principles of Helicopter Aerodynamics*, Cambridge University Press, 2006
[6]	Cheeseman, I.C. and Bennett, W.E., "The Effect of the Ground on a Helicopter Rotor in Forward Flight," ARC R&M 3021, 1957
[7]	Newman, S. et al., "Vortex Ring State Prediction Models," *Aeronautical Journal*, 2001
[8]	Cicolani, L.S. and Kanning, G., "Equations of Motion of Slung-Load Systems," NASA TP-3280, 1992
[9]	NGIC Software Interface Control Document Template
[10]	T-150 UAS AFSIM Plugin Suite – User Acceptance Method (UAM), v3.0
[11]	T-150 UAS AFSIM Plugin Suite – Verification and Validation Plan
	UNCLASSIFIED

9 (U) Acronyms

Table 20: (U) Acronyms

UNCLASSIFIED	
Acronym	Definition
ABI	Application Binary Interface
AER	AFSIM Event Record
AFSIM	Advanced Framework for Simulation, Integration, and Modeling
AGL	Above Ground Level
API	Application Programming Interface
CG	Center of Gravity
CSV	Comma-Separated Values
CUI	Controlled Unclassified Information
DIS	Distributed Interactive Simulation
DLL	Dynamic Link Library
DOF	Degrees of Freedom
ESC	Electronic Speed Controller
HLA	High Level Architecture
ICD	Interface Control Document
MSL	Mean Sea Level
MTOW	Maximum Takeoff Weight
NED	North-East-Down (coordinate frame)
NGIC	National Ground Intelligence Center
OCV	Open-Circuit Voltage
OGE	Out of Ground Effect
RF	Radio Frequency
RPM	Revolutions Per Minute
SOC	State of Charge
UAS	Unmanned Aircraft System
VRS	Vortex Ring State
WSF	Warfighter Simulation Framework
	UNCLASSIFIED

10 (U) Acronyms

Citations